

BCOG 100

Mind and Brain Lecture 2:
Levels of Analysis



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Starting question

Password: three

Question: What are David Marr's Three Levels of analysis.
Give a definition of each in your own words.



Mind-Body Solutions

Two Dimensions of Disagreement

1. **Ontological Dependence:** Is the mind a different kind of “stuff” than the physical, material world?
2. **Explanatory Separability:** Can we understand the physical and the mental independently (whatever they are made of)?

		Ontologically Dependent?	
		Yes	No
Explanatorily Separable	Yes	4	1
	No	2	3

Mind-Body Solutions: Quadrant 4

The Popularity of Quadrant 4:

- Still materialist and scientific (as opposed to quadrant 1 and 3)
- Psychological explanations work
- Theoretical limitations: can you really eliminate all talk of memory, attention, concepts?
- Most sciences are reductive without being eliminativist
- Practical constraints. modeling in terms of neurons is intractable
- Multiple realizability



Mind-Body Solutions: Quadrant 4

Ontologically Dependent?

		Yes	No
Explanatorily Separable	Yes	4	
	No		

Definition: Mind depends on physical systems, but can sometimes be explained in terms that are not reduced to physical ones.

Example: Functionalism

- What makes something a mental state is what it does, not what it's made of. Pain = the state that's caused by injury and leads to avoidance, regardless of whether it's in a brain, a robot, or an alien.
- Mental states are defined by their functions or roles in a system, not their physical substance.

Problem: It struggles with qualia — explaining why these functional roles are accompanied by felt experience (“what it’s like” to be in pain).

Mind-Body Solutions: Quadrant 4

Ontologically Dependent?

		Yes	No
Explanatorily Separable	Yes	4	
	No		

Definition: Mind depends on physical systems, but can sometimes be explained in terms that are not reduced to physical ones.

Example: Emergentism

- Mental phenomena emerge from complex physical systems like the brain, the way liquidity emerges from H₂O molecules.
- The mind is not separate from the brain, but it has new properties that aren't reducible to the properties of neurons.

Problem: Critics ask whether “emergence” really explains anything — or if it’s just a label for our ignorance about how brain activity produces cognition.

Mind-Body Solutions: Quadrant 4

Ontologically Dependent?

		Yes	No
Explanatorily Separable	Yes	4	
	No		

Definition: Mind depends on physical systems, but can sometimes be explained in terms that are not reduced to physical ones.

Example: Multiple Levels of Analysis

- To understand the mind, we must explain it at different levels:
- Each level gives a valid, necessary kind of explanation; no single level captures the whole picture.

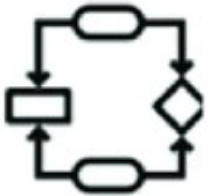
Problem: How do we connect the levels? For example, how do abstract computations (like vision) map onto specific neural processes?



David Marr's Levels of Analysis



Computational level: What is the system doing? What is the problem being solved?



Algorithmic level: How is it being done? What representations and processes are used?

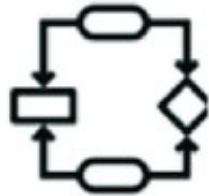


Implementational level: Where and by what physical means is it carried out?

Example: Vision



Computational level: The goal is to recover 3D structure of the world from 2D retinal images.

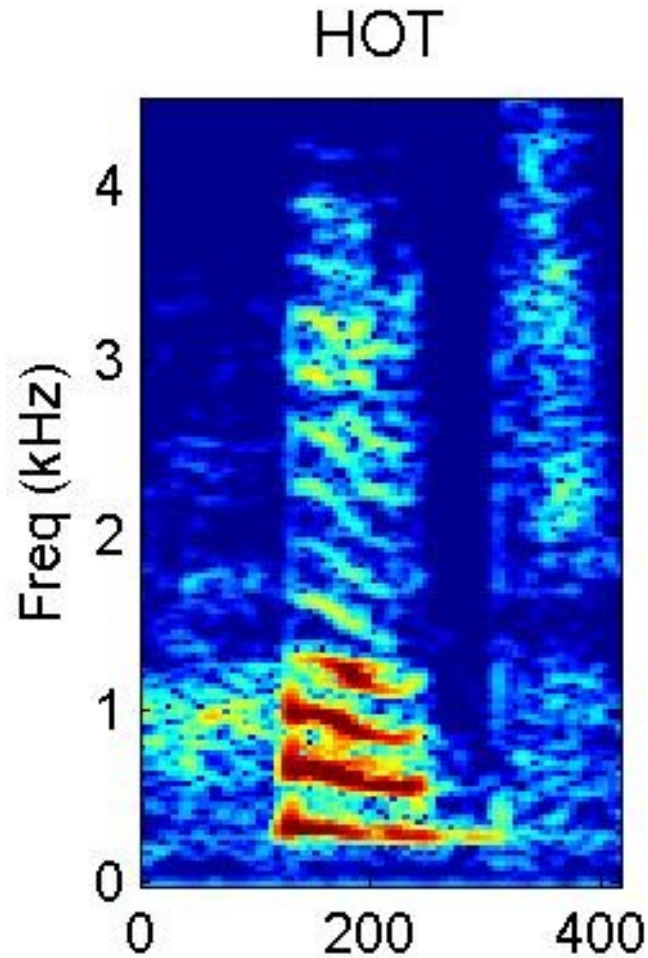


Algorithmic level: Edge detection, depth cues, object segmentation, recognition processes.

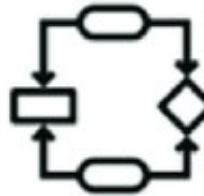


Implementational level: Retinal cells, LGN pathways, cortical neurons in V1, V2, IT, etc.

Example: Language



Computational level: Map an acoustic signal onto a speaker's intended message.



Algorithmic level: Segment words, determine syntactic structure, compute meaning of whole utterance.



Implementational level: Neurons in auditory cortex, temporal lobe, Broca's area, recurrent neural networks, etc.

Strengths and Benefits of Marr's Approach

- Encourages **clarity**: distinguish *what* from *how* from *where*.
- Supports **explanatory separability**: cognitive models don't have to map directly to neurons.
- Helps avoid **reductionist traps** (e.g., dismissing psychology because we have neuroscience).
- Provides a **division of labor** across disciplines (psychology, neuroscience, AI, philosophy).



Challenges to Marr's Approach

- Rigid separation: In practice, levels are not always independent.
- Top-down bias: Assumes the computational problem is well-defined in advance.
- Real systems are messy: Evolution doesn't optimize cleanly for abstract "functions."
- Example: modern AI shows implementational details (architecture, data statistics) shape what algorithms emerge.



Challenges to Marr's Approach

Marr's view:

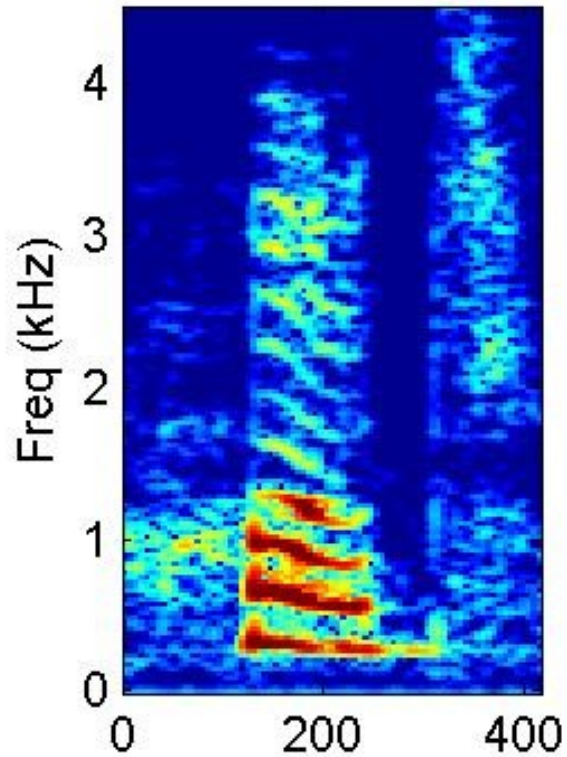
- Computational: The goal is to store experiences and retrieve them later.
- Algorithmic: Processes like encoding, storage, and retrieval.
- Implementational: Brain regions like the hippocampus and cortex.

Problem:

- The way memory works depends heavily on the brain's wiring.
- For example: The hippocampus is especially good at quickly linking events together, but struggles with long-term storage on its own.
- This means the “algorithm” is shaped by the biology — you can't neatly separate the levels.

Another Way of Thinking About Levels

HOT

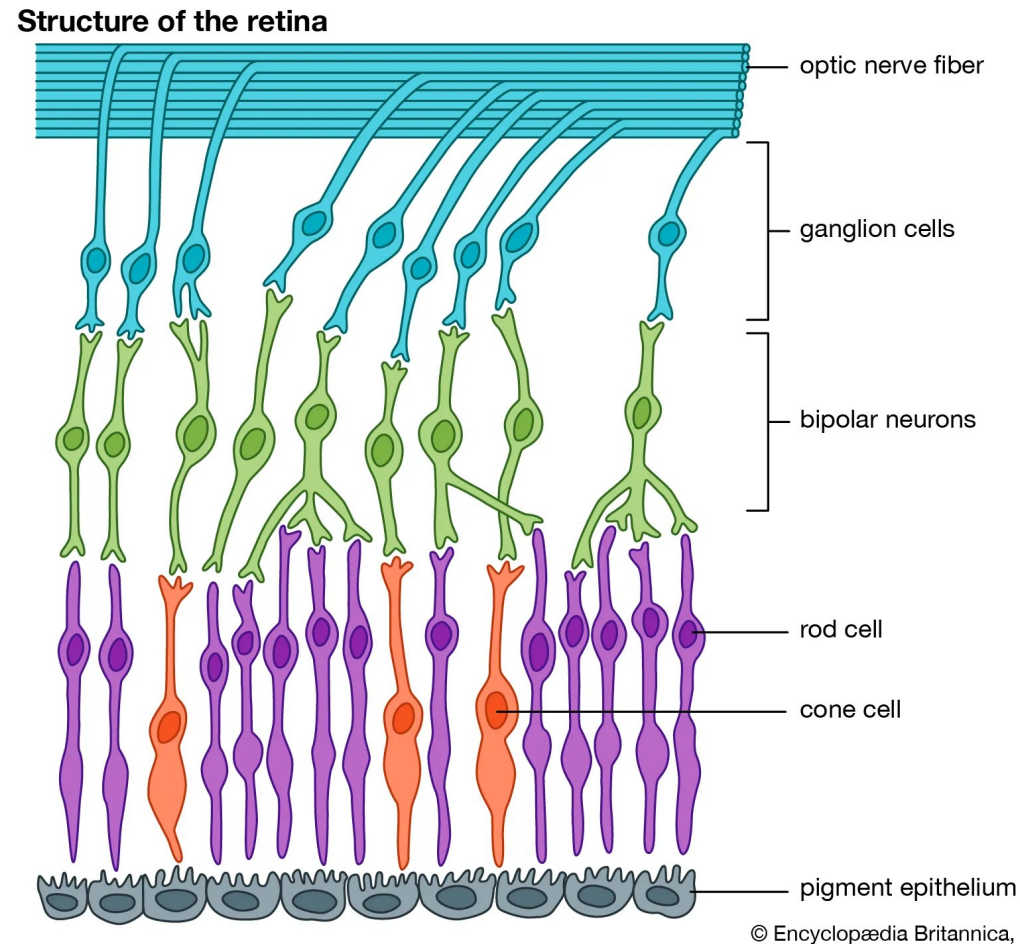


There are a lot of different ways you can move your mouth and say “hot”

There are a lot of different auditory vibrations you can detect and hear “hot”

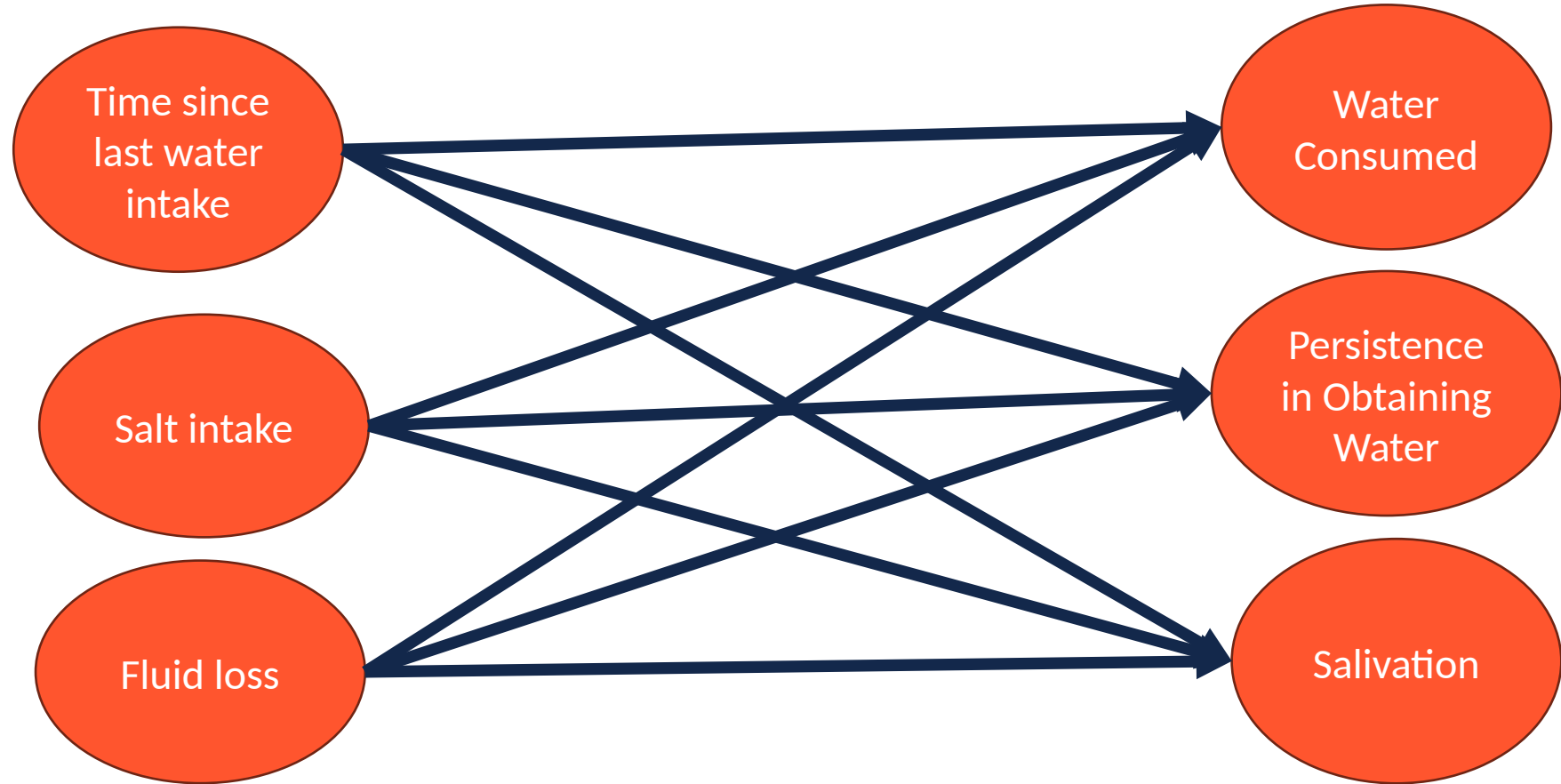
Is speaking or hearing “hot” a “real” thing?

Another Way of Thinking About Levels

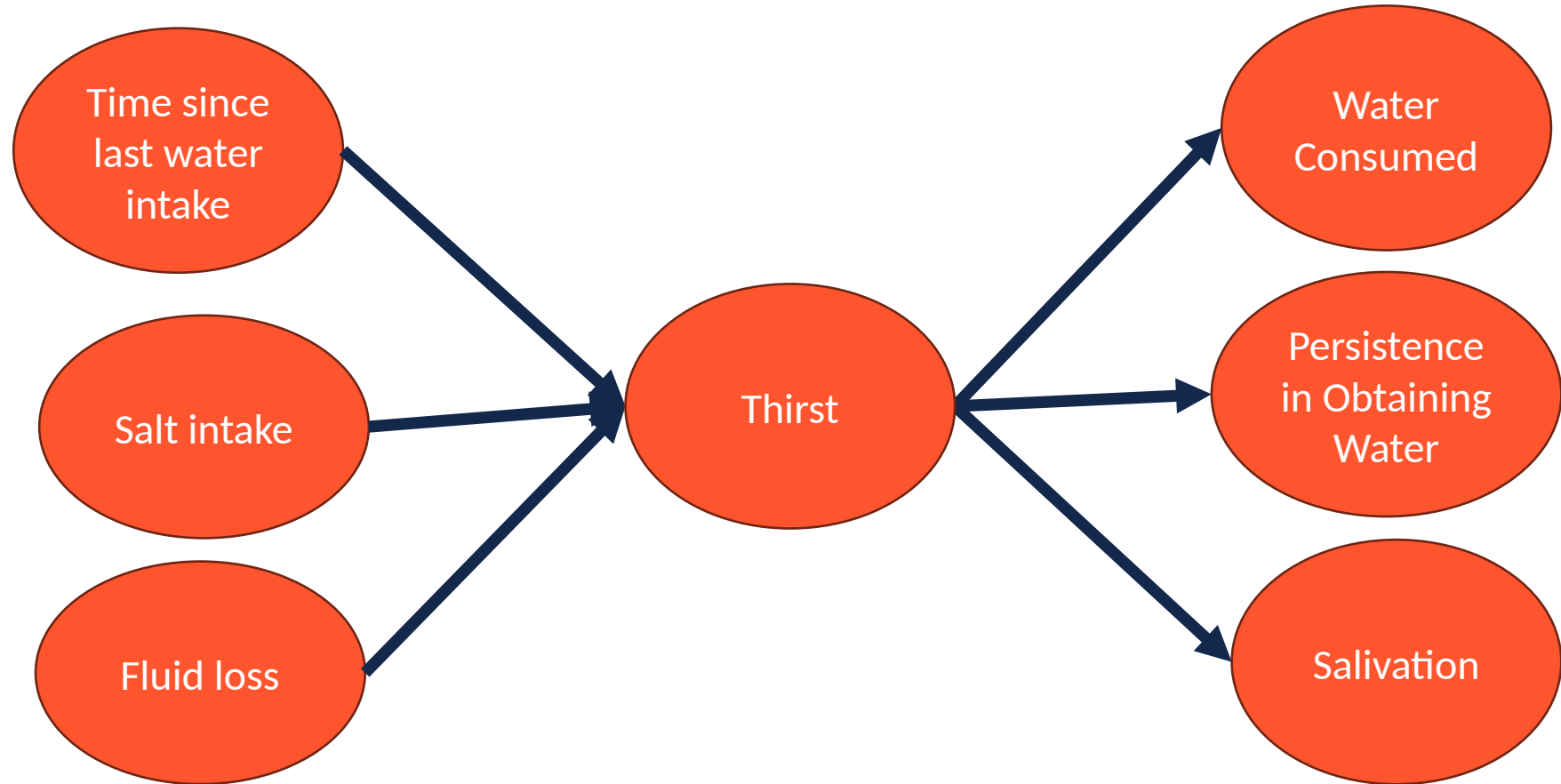


Is the concept useful? Does it lead to predicting things as well, with less complexity?

Another Way of Thinking About Levels

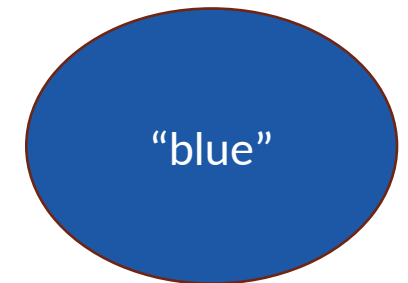
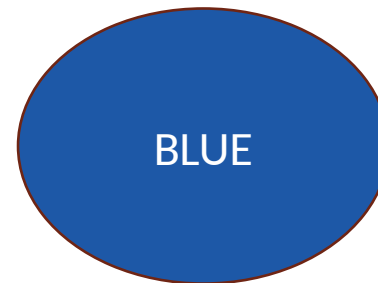
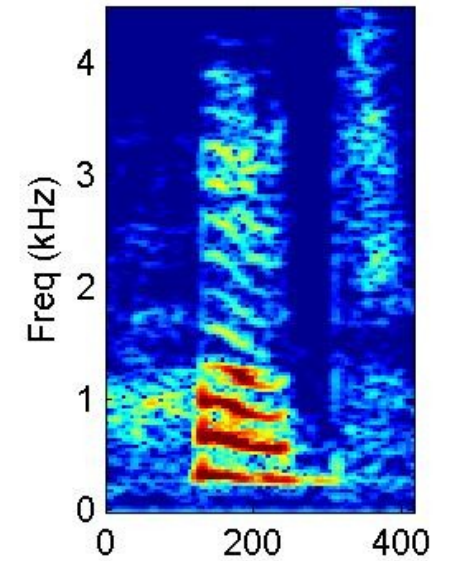
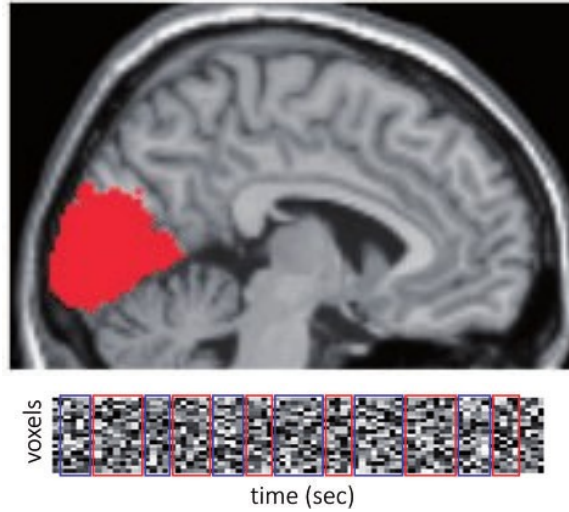
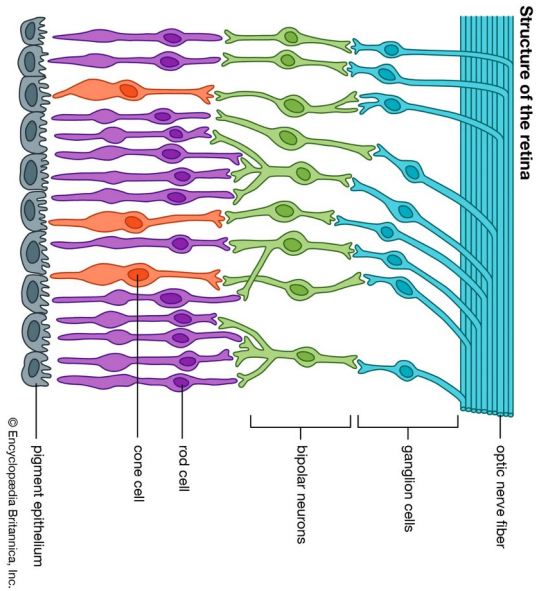


Another Way of Thinking About Levels



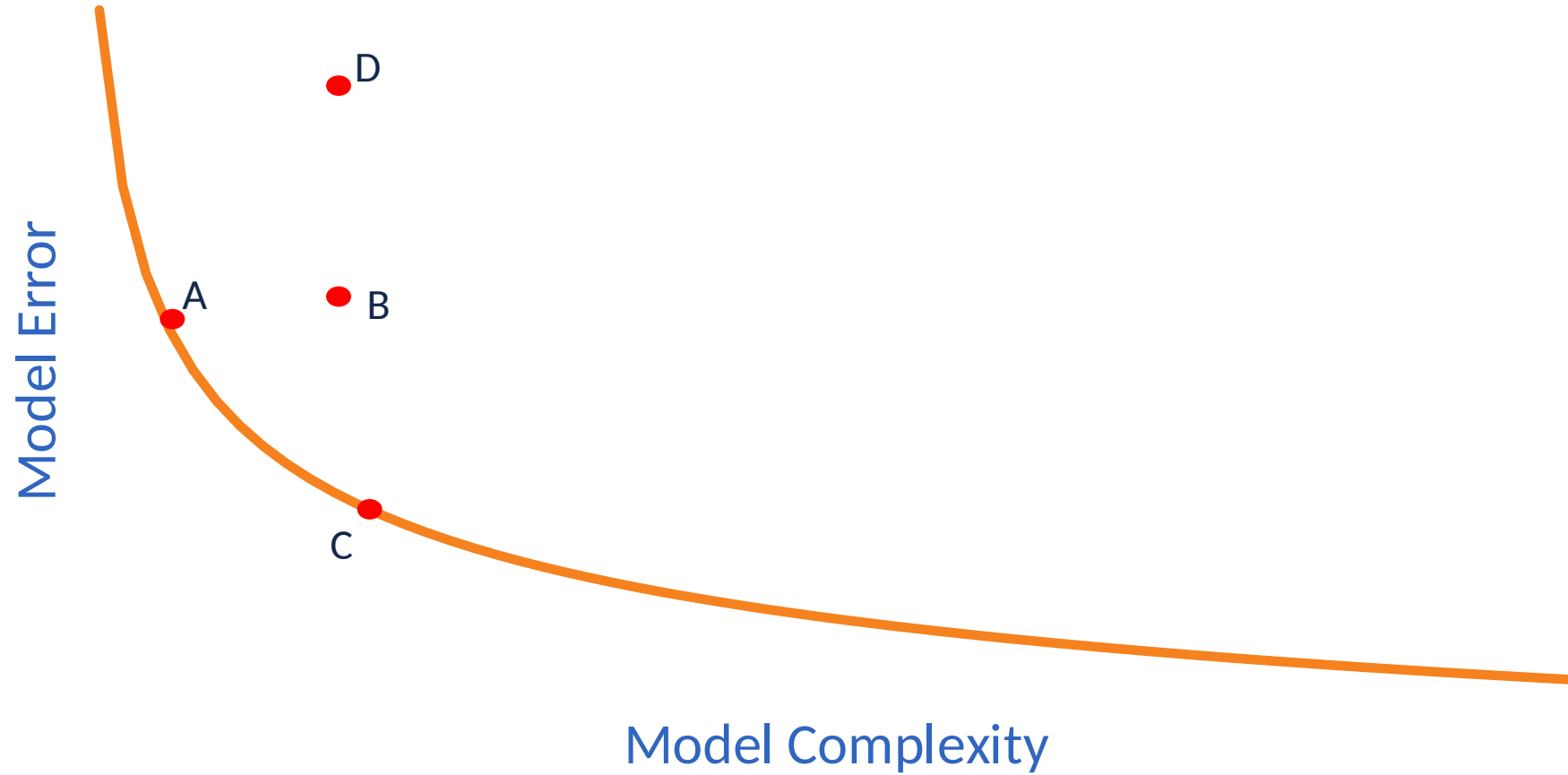
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Another Way of Thinking About Levels



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Another Way of Thinking About Levels



Summary

- Marr's levels remain hugely influential.
- Most cognitive sciences endorse multi-level explanations.
- But modern work highlights interactions between levels: not cleanly separable, often mutually constraining.
- Most scientists are not dogmatic about what the levels are. The focus is more pragmatic.
 - Do your concepts make good predictions
 - Do they help lead to understanding